

# A Unique Maximal Eigenmeasure but an Infinite-Dimensional Bidual Ruelle Eigenspace

Run fa\_banach\_001, lane 17 (agent\_lane\_17)

13 August 2026

**Status: full negative answer, likely valid, pending human review.** For the fair two-symbol Bernoulli shift with zero potential, the maximal eigenmeasure is unique, but the eigenspace of the double transpose on  $L^1(\nu)^{**}$  is infinite-dimensional.

## 1 Exact source question

Cioletti, van Enter, and Ruviaro study the natural extension  $\mathbb{L}_f : L^1(\nu) \rightarrow L^1(\nu)$  of a Ruelle operator. In Section 1.1, arXiv PDF page 4 of [1], they consider the barycenter  $\nu$  of  $\mathcal{G}^*(f)$  and ask about the relation

**Proposed relation in Section 1.1.**

$$\dim_{\mathbb{R}} \mathcal{H} = \# \text{ex}(\mathcal{G}^*(f)), \quad \mathcal{H} = \{\xi \in L^1(\nu)^{**} : \mathbb{L}_f^{**} \xi = \rho(\mathcal{L}_f) \xi\}.$$

The exact passage is reproduced below.

are based on the general theory of linear operators. We shall remark that our approach does not need the expansivity hypothesis, sometimes considered in the literature on Thermodynamic Formalism.

**1.1. Directions for future research.** One of our main results, Theorem 4.1 states that for any continuous potential  $f$ , and  $\nu \in \mathcal{G}^*(f)$ , there is a positive element  $\xi_f \in L^1(\nu)^{**}$  which determines a  $\sigma$ -invariant Borel finitely additive measure given by  $A \mapsto \xi_f(\ell_A) \equiv \mu(A)$ , where  $\ell_A$  denotes the bounded linear functional on  $L^1(\nu)$  given by  $\ell_A(\varphi) = \int_X 1_A \varphi d\nu$ .

By Theorem 3.1 in [59] we have

$$L^1(\nu)^{**} \cong \{\gamma : \mathcal{B}(X) \rightarrow \mathbb{R} : \gamma \in \text{ba}(\mathcal{B}(X)) \text{ satisfying } \nu(Z)=0 \implies \gamma(Z)=0\},$$

where  $\text{ba}(\mathcal{B}(X))$  stands for the space of all Borel bounded finitely additive signed measures. By Theorem 3.1 in [59], we can identify the finitely additive measure  $\mu$ , constructed above, with a bounded linear functional  $F_\mu : L^\infty(\nu) \rightarrow \mathbb{R}$ . The restriction of  $F_\mu$  to  $C(X)$  determines a countably additive  $\sigma$ -invariant measure, because of the Riesz-Markov Theorem. It would be very interesting to know whether such a Borel probability measure is actually an equilibrium state associated to  $f$ , and how to determine the support of such a measure in terms of its Yosida-Hewitt decomposition  $\mu = \mu_a + \mu_c$ , see [59, 66] for details on this decomposition.

Another interesting question related to Theorem 4.1 is the following. Consider  $\nu$  the barycenter of the convex set  $\mathcal{G}^*(f)$  and let  $\mathcal{H} \equiv \{\xi \in L^1(\nu)^{**} : \mathbb{L}_f^{**} \xi = \rho(\mathcal{L}_f) \xi\}$ . Some of the results in [19] suggest that  $\dim_{\mathbb{R}} \mathcal{H} = \# \text{ex}(\mathcal{G}^*(f))$  (the set of extreme points in  $\mathcal{G}^*(f)$ ). If such relation indeed holds, it has as a consequence a new criterion for the existence of phase transitions for one-dimensional one-sided lattice statistical mechanical spin systems, which in turn would reduce this problem to a problem of spectral analysis of Markov operators.

Yet another question, which seems to be more puzzling, is the following. When the Yosida-Hewitt decomposition of  $\mu$  is trivial, meaning that  $\mu = \mu_a$  is purely finitely additive, can any statistics along typical orbits of observables be obtained,

We give a counterexample in the simplest possible symbolic system. In fact, the left-hand side is infinite while the right-hand side equals one.

## 2 The zero-potential Bernoulli system

Let  $E = \{0, 1\}$ , let  $p = \frac{1}{2}(\delta_0 + \delta_1)$ , put  $X = E^{\mathbb{N}}$ , and write  $\sigma : X \rightarrow X$  for the left shift. Take  $f = 0$ . Then

$$(\mathcal{L}\varphi)(x) = \frac{1}{2}\varphi(0x) + \frac{1}{2}\varphi(1x), \quad \varphi \in C(X), \quad (1)$$

and  $\rho(\mathcal{L}) = 1$ .

**Lemma 1.** *The set of maximal eigenmeasures is the singleton*

$$\mathcal{G}^*(0) = \{\nu\}, \quad \nu = \left(\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1\right)^{\mathbb{N}}.$$

Consequently  $\# \text{ex}(\mathcal{G}^*(0)) = 1$ , and its barycenter is  $\nu$ .

*Proof.* Since  $\mathcal{L}1 = 1$  and  $\|\mathcal{L}\| = 1$ , the spectral radius is one. If a Borel probability measure  $m$  satisfies  $\mathcal{L}^*m = m$ , then for every binary word  $w = w_1 \dots w_n$ , whose cylinder is denoted  $[w]$ , equation (1) gives

$$\mathcal{L}1_{[w]} = \frac{1}{2}1_{[w_2 \dots w_n]}.$$

(For  $n = 1$ , the last indicator is 1.) Hence  $m([w]) = 2^{-1}m([w_2 \dots w_n]) = 2^{-n}$ . Cylinder probabilities determine the measure, so  $m = \nu$ . Conversely, the product measure plainly is fixed.  $\square$

Let  $\mathbb{L} : L^1(\nu) \rightarrow L^1(\nu)$  be the natural extension of (1). It is a positive contraction. Under the identification  $L^1(\nu)^* = L^\infty(\nu)$ , its adjoint is the Koopman isometry

$$U\psi = \psi \circ \sigma. \quad (2)$$

Indeed, first for bounded measurable functions and then by density,

$$\int_X (\mathbb{L}h)\psi \, d\nu = \int_X h(\psi \circ \sigma) \, d\nu.$$

Thus  $\mathbb{L}^{**} = U^*$  on  $L^1(\nu)^{**} = L^\infty(\nu)^*$ .

## 3 Periodic-orbit states in the bidual

Fix a periodic point  $y \in X$ . Let  $C_n(y)$  be the cylinder consisting of sequences whose first  $n$  coordinates agree with those of  $y$ . Since  $\nu(C_n(y)) = 2^{-n}$ , define a positive norm-one functional on  $L^\infty(\nu)$  by

$$\lambda_n^y(\psi) = 2^n \int_{C_n(y)} \psi \, d\nu. \quad (3)$$

Banach–Alaoglu gives a weak-star cluster point  $\lambda_y$  of this net in  $L^\infty(\nu)^*$ . If  $\varphi \in C(X)$ , uniform continuity and the fact that the diameters of  $C_n(y)$  tend to zero imply

$$\lambda_y(\varphi) = \varphi(y). \quad (4)$$

Thus  $\lambda_y$  is an  $L^\infty(\nu)^*$ -state extending evaluation at the  $\nu$ -null point  $y$  on the continuous subspace.

Now form the Cesàro averages

$$A_N^y = \frac{1}{N} \sum_{j=0}^{N-1} (U^*)^j \lambda_y. \quad (5)$$

They are again positive norm-one functionals. Take a weak-star cluster point  $\eta_y$ . Since

$$\|U^* A_N^y - A_N^y\| \leq \frac{\|(U^*)^N \lambda_y\| + \|\lambda_y\|}{N} \leq \frac{2}{N},$$

we have  $U^* \eta_y = \eta_y$ , so  $\eta_y \in \mathcal{H}$ .

If the orbit  $O(y)$  has period  $r$ , equations (2), (4), and (5) show that on  $C(X)$ ,

$$\eta_y(\varphi) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=0}^{N-1} \varphi(\sigma^j y) = \frac{1}{r} \sum_{z \in O(y)} \varphi(z). \quad (6)$$

Thus the restriction of  $\eta_y$  is precisely the uniform probability measure on the periodic orbit.

**Theorem 1.** *For the system above,  $\mathcal{H}$  is infinite-dimensional even though  $\#\text{ex}(\mathcal{G}^*(0)) = 1$ . Hence the proposed dimension relation is false.*

*Proof.* For  $m \geq 1$ , let  $y^{(m)} = (10^m)^\infty$ . These points belong to pairwise distinct periodic orbits  $O_m$ . Construct  $\eta_m := \eta_{y^{(m)}}$  as above. Suppose that a finite linear combination  $\sum_{m \in F} a_m \eta_m$  vanishes. For any fixed  $k \in F$ , the finite closed sets  $O_k$  and  $\bigcup_{m \in F \setminus \{k\}} O_m$  are disjoint, so there is a continuous function  $\varphi_k$  equal to one on  $O_k$  and zero on all the other orbits. Restricting the relation to  $C(X)$  and using (6) gives  $a_k = 0$ . Hence the family  $(\eta_m)_{m \geq 1}$  is linearly independent in  $\mathcal{H}$ .

Lemma 1 gives the claimed value one on the right-hand side. □

## 4 Verification, scope, and novelty

The construction uses only positive norm-one functionals. The first weak-star limit extends point evaluation from  $C(X)$  to  $L^\infty(\nu)$ ; the second turns it into an invariant state. Norm convergence of the coboundary in (5) makes invariance independent of the chosen subnet. Linear independence is certified after restriction to  $C(X)$ , where the states are ordinary mutually singular periodic-orbit measures.

This does not contradict uniqueness in Lemma 1: the additional states are finitely additive elements of  $L^1(\nu)^{**}$  and do not come from  $L^1(\nu)$ . Thus a modified question restricted to the canonical  $L^1(\nu)$  copy could behave differently.

The bounded novelty search checked the run's solution, attempt, proof-gap, and ledger indexes; the parsed local arXiv corpus; exact-title and exact-question searches; and focused arXiv/web searches for the bidual eigenspace dimension and finitely additive Ruelle eigenvectors. It found the source's 2023 publication and neighboring work on eigendistributions, but no later explicit resolution of this dimension relation. The invariant-mean mechanism itself is classical in spirit, so novelty is plausible rather than certified. Human review should prioritize the identification  $\mathbb{L}^* = U$  and the two weak-star cluster-point passages.

## References

- [1] L. Cioletti, A. van Enter, and R. Ruviano, *The Double Transpose of the Ruelle Operator*, *Monatsh. Math.* **200** (2023), 523–544; arXiv:1710.03841v4.